

Atomic Structure and the Bohr Model

Rutherford's Nuclear Model

According to Rutherford's nuclear model, an atom consists of a small, dense, positively charged nucleus surrounded by electrons moving around it, with most of the atom being empty space.

This model could not explain why electrons do not spiral into the nucleus, nor could it explain the discrete line spectrum observed for the hydrogen atom.

Bohr's Postulates

Bohr proposed that an electron in an atom can only occupy certain fixed, quantized energy levels, and while in one of these levels it does not radiate energy.

The energy of an electron in the n th level of a hydrogen atom is given by the formula $E_n = -R_H / n^2$, where R_H is the Rydberg constant, approximately 2.179×10^{-18} joules, and n is the principal quantum number, a positive integer.

An electron can move between energy levels by absorbing or emitting a photon whose energy equals the difference between the two energy levels: the energy of the emitted photon equals E_{initial} minus E_{final} for a transition to a lower level.

Limitations

The Bohr model successfully explains the hydrogen emission spectrum but does not accurately predict the spectra of multi-electron atoms, and was later superseded by the quantum mechanical model of the atom.